TASK 21 | FINAL RESEARCH REPORT

MATLAB Radar Analysis, Data Quality, Temporal Behavior and CPU Benchmarking

Project: Uncertainty-Constrained Energy-Aware ARM-FPGA Hardware-Software Co-Design for Tri-Sensor Perception

Evidence: Task 20 radar CSV exports and Task 21 MATLAB/Python results | Status: Analysis and software benchmarks completed; hardware energy and radar object-detection quality unverified.

Critical result	Observed outcome
Corrected 50 m dataset	5,696 candidate reflections, including 82 at frame 0
Analysis dataset	5,614 returns across frames 1-99 after excluding frame zero
Extended range	Maximum measured range 49.9283 m; 2,655 returns beyond 10 m
Fastest measured software implementation	MATLAB frame-by-frame aggregation, median 0.025771 ms per 99-frame pass

1. Research question, scope and methodology

Can the corrected GEMSTAR radar recordings be converted into reproducible MATLAB datasets, checked for invalid observations, characterized over range and frames, and processed efficiently in software before considering ARM-FPGA acceleration? Task 21 extends the recorded-data foundation from Task 20; it does not independently establish real-world radar detection accuracy or electrical energy savings.

2. Workspace and traceable inputs | Task 21.1

Created tests/Task_21_Final_Research with separate directories for MATLAB/Python code, imported data, graphs, CSV results, benchmarks, findings, research reports, configurations and raw-data references. Saved a source inventory and research setup notes. Task 20 source files were treated as immutable inputs.

3. MATLAB dataset import | Task 21.2

Task 20 dataset	Rows	Columns
radar_world_points	11,825	15
50m_radar_returns	5,696	9
ray_alignment	5	13
ray_box_intersections	10	10

MATLAB R2026b imported all four CSV datasets using readtable and stored the resulting tables in MAT files. The source path was preserved alongside the imported table.

4. Data-quality validation | Tasks 21.3 and 21.3B

The first quality-check script could not consistently count missing fields, so a corrected audit separately examined NaN, infinity and blank textual fields. This correction avoided interpreting skipped missing-value checks as zero missing entries.

Quality check	Verified result
50 m candidate returns across frames 0-99	5,696
Frame-zero returns	82
Near-zero returns (<1 mm), all in frame zero	50
Returns after excluding frame zero	5,614
Missing entries in 5-row ray-alignment table	5 (unavailable nearest-ray ranges)
Missing entries in 10-row ray-box table	15 (absent entries / non-intersections)
Infinite values in four audited datasets	0 reported

Empty range fields are expected where no positive ray return is present; they must not be imputed as zero-distance object detections. The archived world-points table uses a different or unrecognized frame-column name: an unknown frame count is not evidence of invalid measurements.

5. Range-distribution analysis | Task 21.4

Compared an archived original 10 m dataset (11,825 returns from multiple recordings) against a corrected 50 m recording (5,614 returns after excluding frame zero). The original maximum was 9.9897 m; the corrected maximum was 49.9283 m. In the corrected recording, 2,655 returns exceeded 10 m, 1,390 exceeded 20 m and 626 exceeded 40 m. These distributions are descriptive because the recordings are not scene-matched.

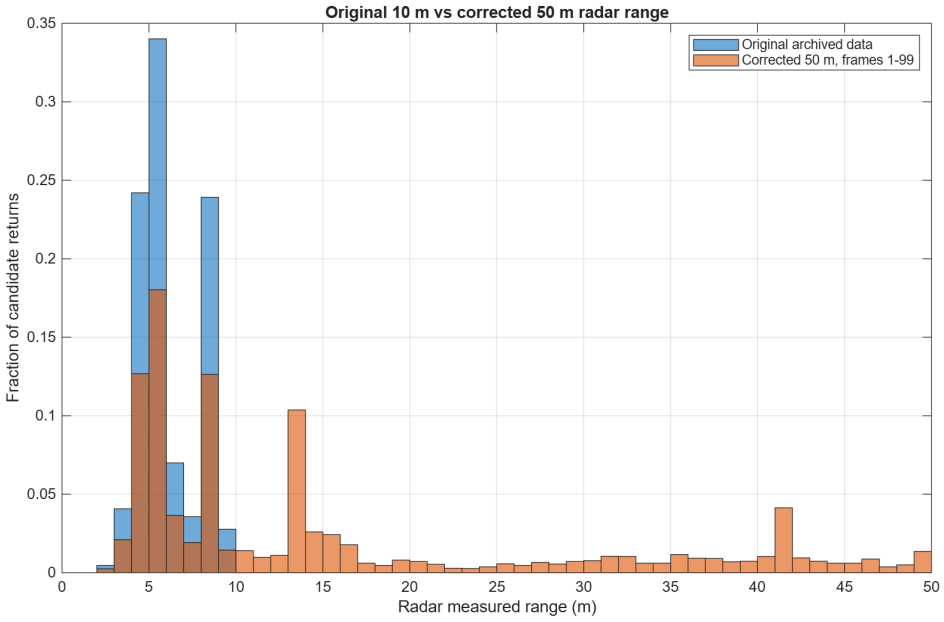


Figure 1. Original archived 10 m and corrected 50 m range distributions (user-supplied MATLAB figure).

6. Radar signal-strength and near-range behavior | Task 21.4

The recorded simulator signal-strength values decrease as the range increases in this dataset. This is a characterization of the GEMSTAR radar model output, not a calibrated physical signal-to-noise ratio or proof of object-detection performance.

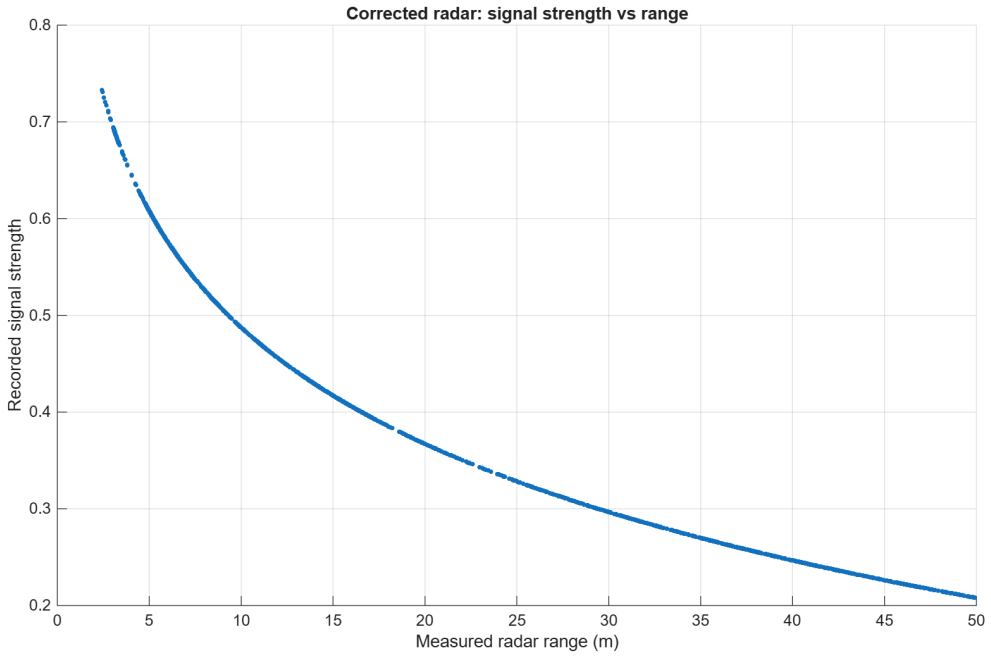


Figure 2. Recorded radar signal strength versus measured range (MATLAB).

The next analysis compares measurements in the shared 0-10 m interval. Both samples have median 5.9114 m within that interval. Similar histogram shapes cannot prove equivalence because data were drawn from different recordings and scenarios.

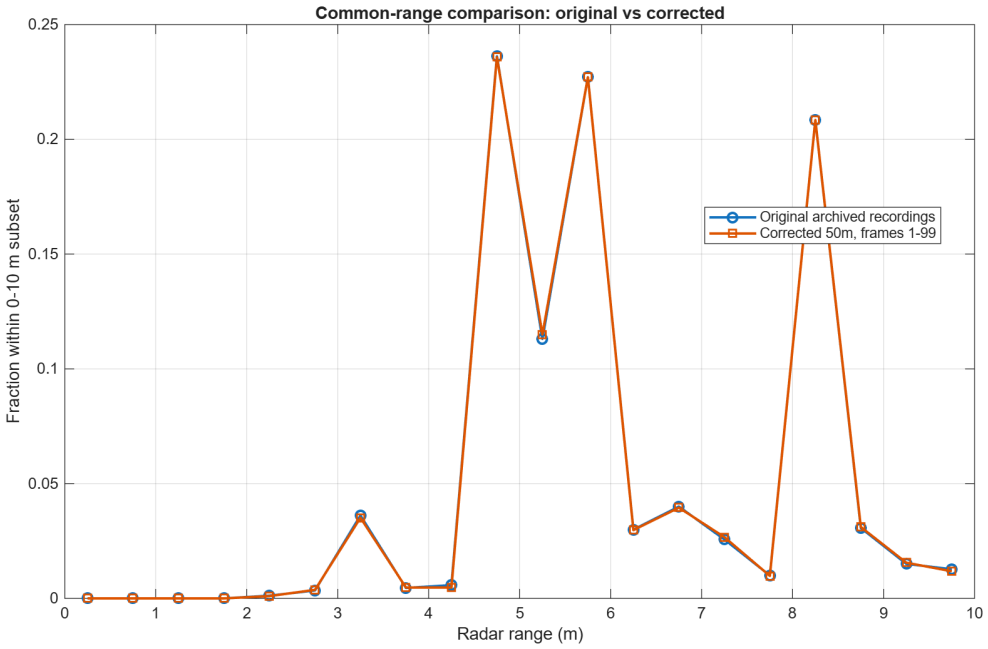


Figure 3. Normalized common-range (0-10 m) comparison; unpaired recordings (MATLAB).

7. Corrected 50 m range-band investigation | Task 21.5

Range band	Candidate returns	Share of corrected returns
0-10 m	2,959	52.708%
10-20 m	1,265	22.533%
20-30 m	288	5.130%
30-40 m	476	8.479%
40-50 m	626	11.151%
Total	5,614	100% (rounding)

The corrected recording includes 2,655 returns beyond 10 m (47.29%). This verifies the presence of recorded ranges that the earlier 10 m configuration did not generate. Counts alone are not an estimate of sensitivity, false-positive rate or detection recall.

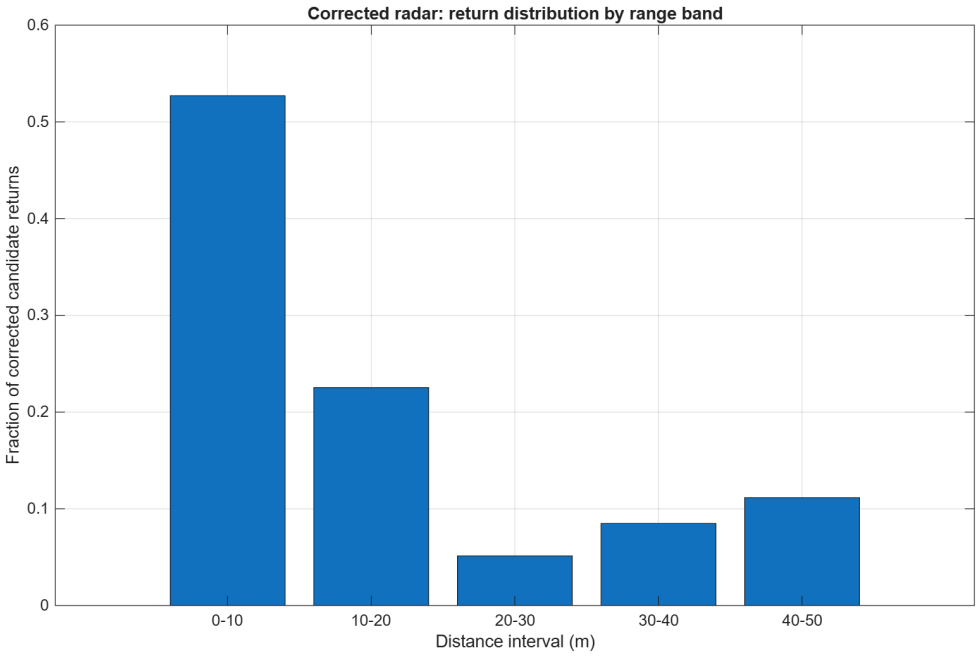


Figure 4. Distribution of corrected candidate returns across five 10 m distance bands.

8. Interpretation of the revised range configuration

Runtime logging and return-distance analysis in Task 20 had already supported the successful configurable 50 m radar correction. Task 21 adds independent MATLAB-side descriptive analysis of those recorded values. A controlled A/B test would require paired scenes with equivalent actors, motion and capture timing.

9. Radar temporal analysis | Task 21.6

Calculated per-frame return counts, returns beyond 10 m, median and maximum range, and signal summaries across frames 1-99. Observed 5,614 returns, between 53 and 63 returns per frame (mean 56.7071), no frames with zero candidate returns, and no count changes exceeding the exploratory 10-return jump threshold. Frame indices represent sample ordering, not wall-clock latency.

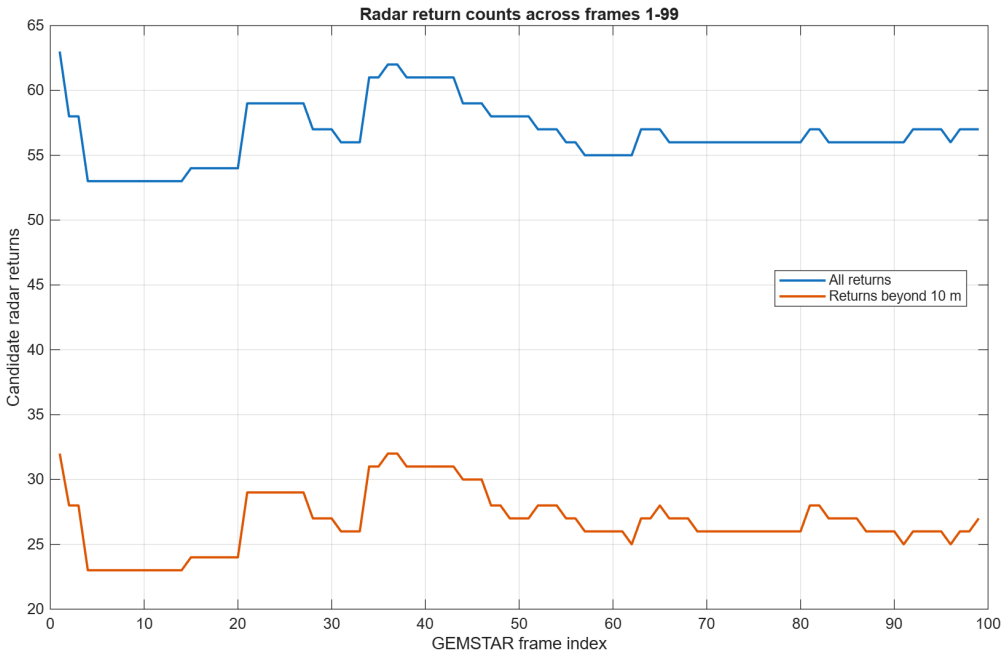


Figure 5. Total candidate returns and returns beyond 10 m for each frame (MATLAB).

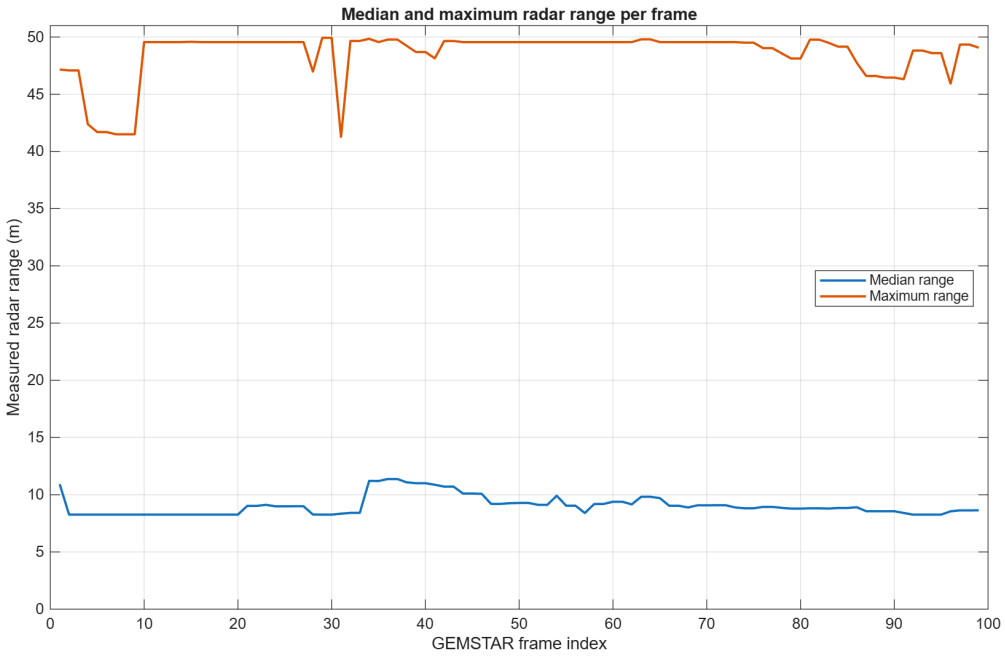


Figure 6. Median and maximum measured radar range by frame (MATLAB).

10. Temporal change and stability interpretation | Task 21.6

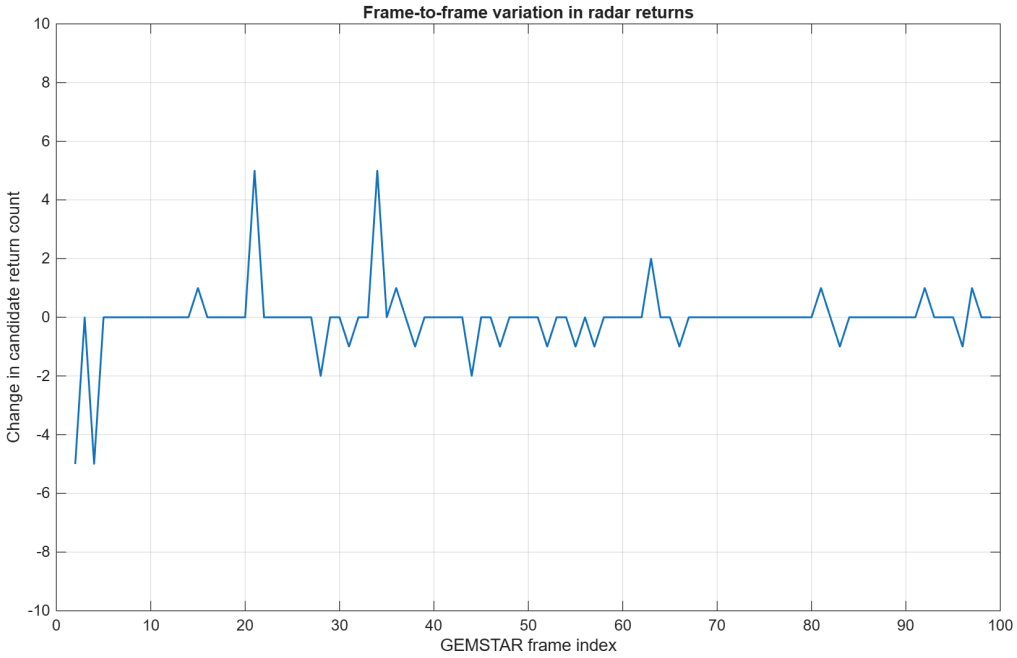


Figure 7. Frame-to-frame changes in candidate return counts (MATLAB).

The count changes are limited within the selected threshold in this recording, supporting a narrow statement of stable return counts. This is not enough to establish correct obstacle detection, reliable object association or robustness under fog and dense traffic.

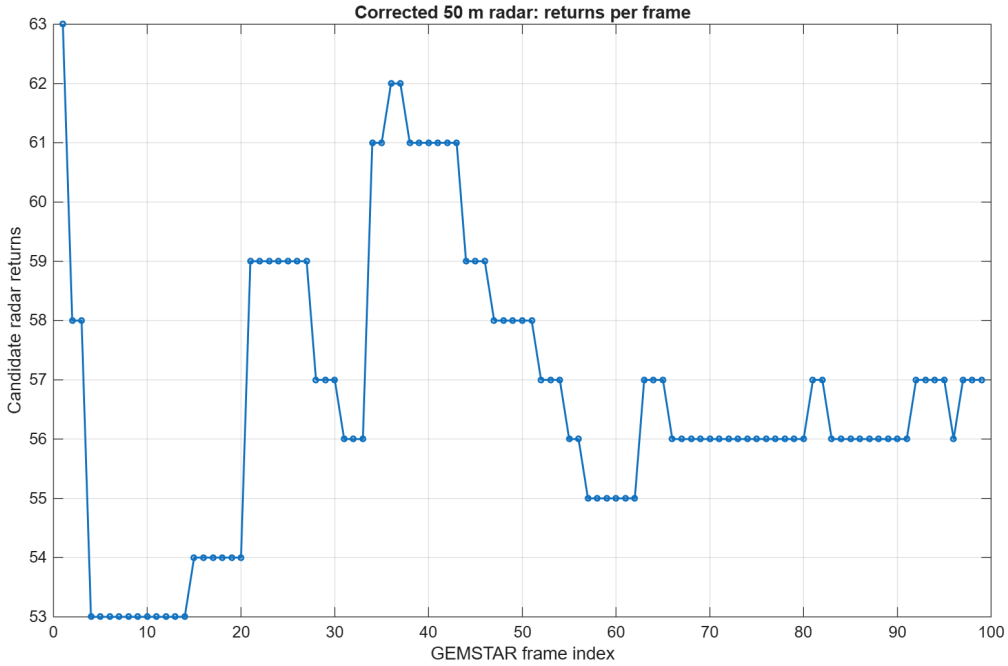


Figure 8. Candidate returns per frame from the corrected radar recording (MATLAB).

11. MATLAB desktop CPU benchmark | Task 21.7

Used five timed trials and 100 repetitions per trial for the same 99-frame aggregation workload. Both MATLAB implementations passed functional checks on valid-return count, >10 m counts and per-frame range sums. Timing excluded MAT loading, disk I/O, OSI decoding and transfer to a hardware accelerator.

Implementation	Median (ms/pass)	Minimum	Maximum
Frame-by-frame MATLAB	0.025771	0.025755	0.026812
Vectorized MATLAB	0.065087	0.064618	0.068622

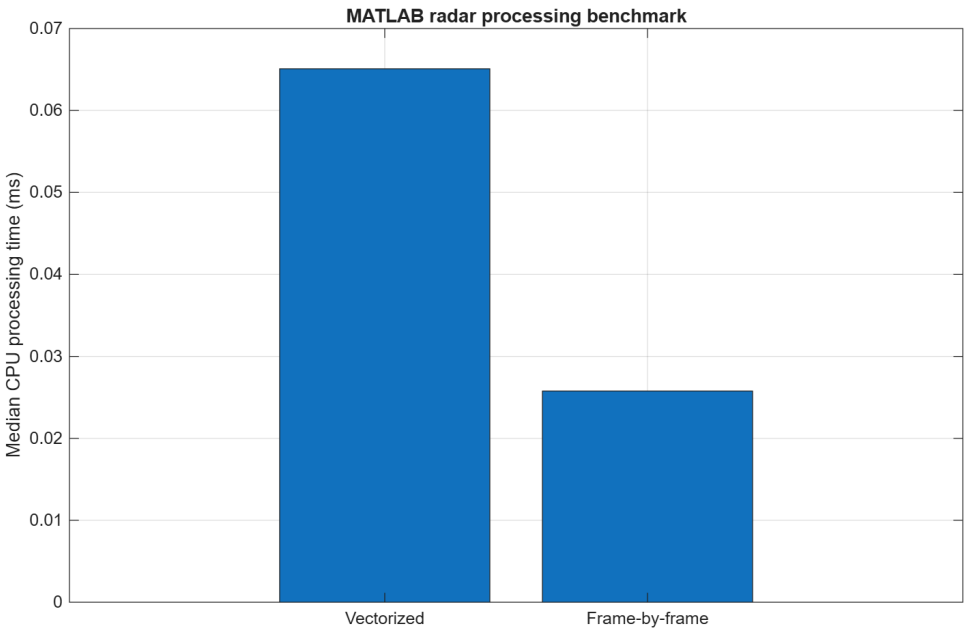


Figure 9. MATLAB radar aggregation execution-time comparison.

12. Python/NumPy comparison | Task 21.8

On the same Windows desktop, Python 3.11.9 / NumPy 2.4.6 performed the corresponding filtering and frame aggregation. Both Python variants passed result-consistency checks: 5,614 accepted returns and 2,655 beyond 10 m. The trials are exploratory because language runtime, implementation details, CPU frequency and timing frameworks differ.

13. Combined cross-language benchmark | Tasks 21.8-21.9

Rank	Implementation	Median ms / 99-frame pass
1	Frame-by-frame MATLAB	0.025771
2	NumPy vectorized	0.051844
3	Vectorized MATLAB	0.065087
4	Python loop	10.251578

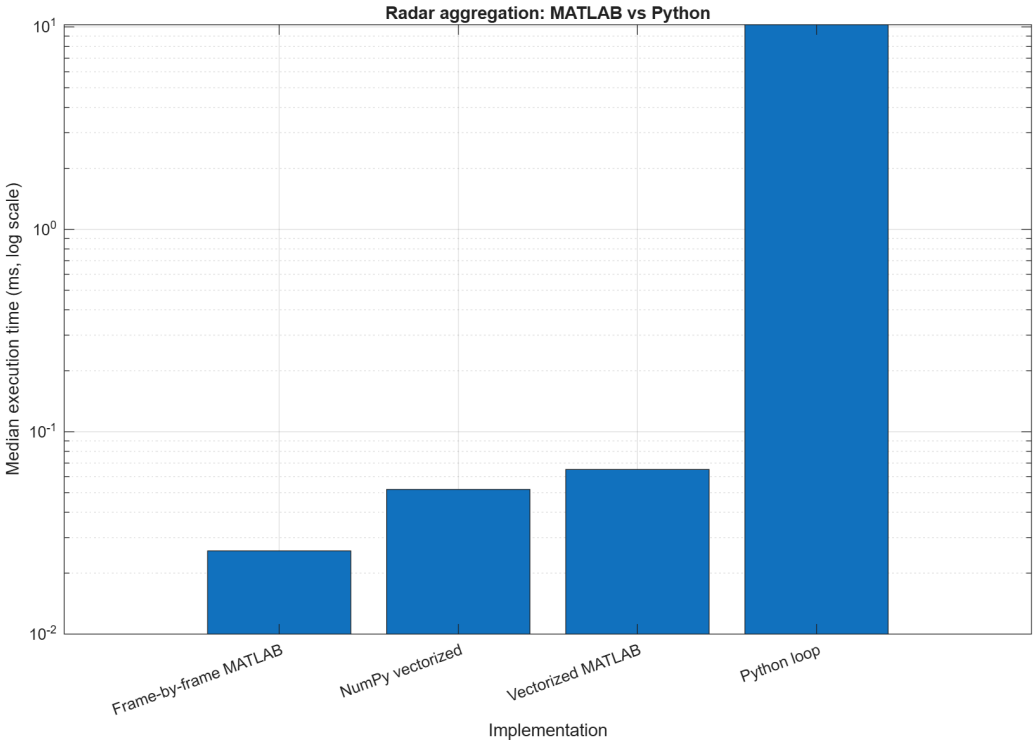


Figure 10. MATLAB and Python desktop kernel execution times; logarithmic time axis.

Interpretation: basic radar aggregation is already a very small CPU workload. Transferring the same data into FPGA fabric may cost more time and energy than this computation saves. A future ARM board benchmark must evaluate compute, DMA, memory, scheduling and idle/static power together.

14. Hardware acceleration screening | Task 21.9

Workload	Preliminary direction	Evidence needed
Radar aggregation	Low initial FPGA priority	ARM runtime, transfer and energy overhead
LiDAR point filtering	FPGA candidate	Profiling at realistic point counts
LiDAR voxel reduction	FPGA candidate	Memory-access and precision analysis
Camera spatial filtering	FPGA candidate	Pixel throughput, line buffers and quality
Sensor fusion / EKF	CPU baseline initially	Validated associations and deadline performance

The candidate assessment is a design hypothesis, not a hardware result. CPU-time savings do not imply lower energy; total board power, accelerator startup, DMA traffic, static power and throughput must be measured.

15. Final audit and evidence archive | Task 21.10

Completed the nine-directory research structure, source inventory, MATLAB/Python scripts, MAT imports, graphs, CSV statistics, timing records and findings. Fourteen critical files were checked during Task 21.10B before archive generation. A temporary ZIP was later deleted at the user's request to keep one visible research folder per task.

Research directory	Purpose
01_MATLAB_Code	MATLAB scripts and Python comparison benchmark
02_Imported_Data	Imported MAT tables with source paths
03_Graphs	MATLAB analysis and benchmark PNG figures
04_CSV_Results	Quality, range-band and temporal data
05_Benchmark_Results	Trial timings, comparison summary, environment records
06_Research_Findings	Audit notes and documented conclusions
07_Research_Report	This final report PDF
08_Configurations	Configuration evidence when applicable
09_Raw_Data_References	Source inventory and file manifest

16. Scientific limitations and claims boundary

- The original 10 m and corrected 50 m archives are not paired controlled experiments; differences in return counts cannot be assigned solely to the radar range setting.
- The simulator signal-strength curve is not calibrated SNR and cannot substitute for obstacle-detection precision or recall.
- Target association, sensor coordinate transforms, ray-to-collision identification and acquisition-time equivalence remain unresolved from Task 20.
- Frame-zero contains 82 candidate observations including 50 near-zero returns; all frame-zero observations were excluded from the corrected Task 21 frame analysis.
- Measured timing is a very small kernel-level desktop CPU experiment; it excludes sensor decoding, I/O, queues, fusion, transport and hardware transfers.
- No electrical board power or energy was measured, and no ARM-FPGA improvement has been established.

17. Overall conclusions and next research phase

Task 21 completed a documented and reproducible MATLAB-centered analysis of the corrected GEMSTAR radar data. It established a consistent filtered dataset of 5,614 candidate observations across 99 frames, described the extension to 49.9283 m, characterized return-count variability and measured desktop kernel execution times. The fastest measured aggregation approach was frame-by-frame MATLAB at 0.025771 ms per dataset pass, demonstrating that simple radar aggregation is not an obvious first FPGA acceleration target.

For Task 22, priority should shift to reusable, scenario-independent OSI ingestion and synchronization, testing distinct maps, fog levels and traffic conditions. Before Task 24 sensor fusion, verify camera and LiDAR data and coordinate transformations. Actual energy-aware ARM-FPGA partitioning requires instrumented on-board power and latency measurements.

18. Reproducibility notes

Recorded 50 m source: 20261102T180104Z_sv_380_X_806_GEMSTAR.osi (Task 20 output). MATLAB: R2026b (26.2.0.3386108). Python: 3.11.9, NumPy 2.4.6. Desktop processor identifier: AMD64 Family 26 Model 96 Stepping 0, AuthenticAMD. Archive source paths and per-script parameters are retained in Task_21_Final_Research; the report summarizes user-reported outputs and supplied original figures without re-running the source datasets.

Final status: Task 21 MATLAB analysis, benchmarking and evidence organization completed. Sensor-detection accuracy, hardware energy savings and final ARM-FPGA implementation are future milestones.